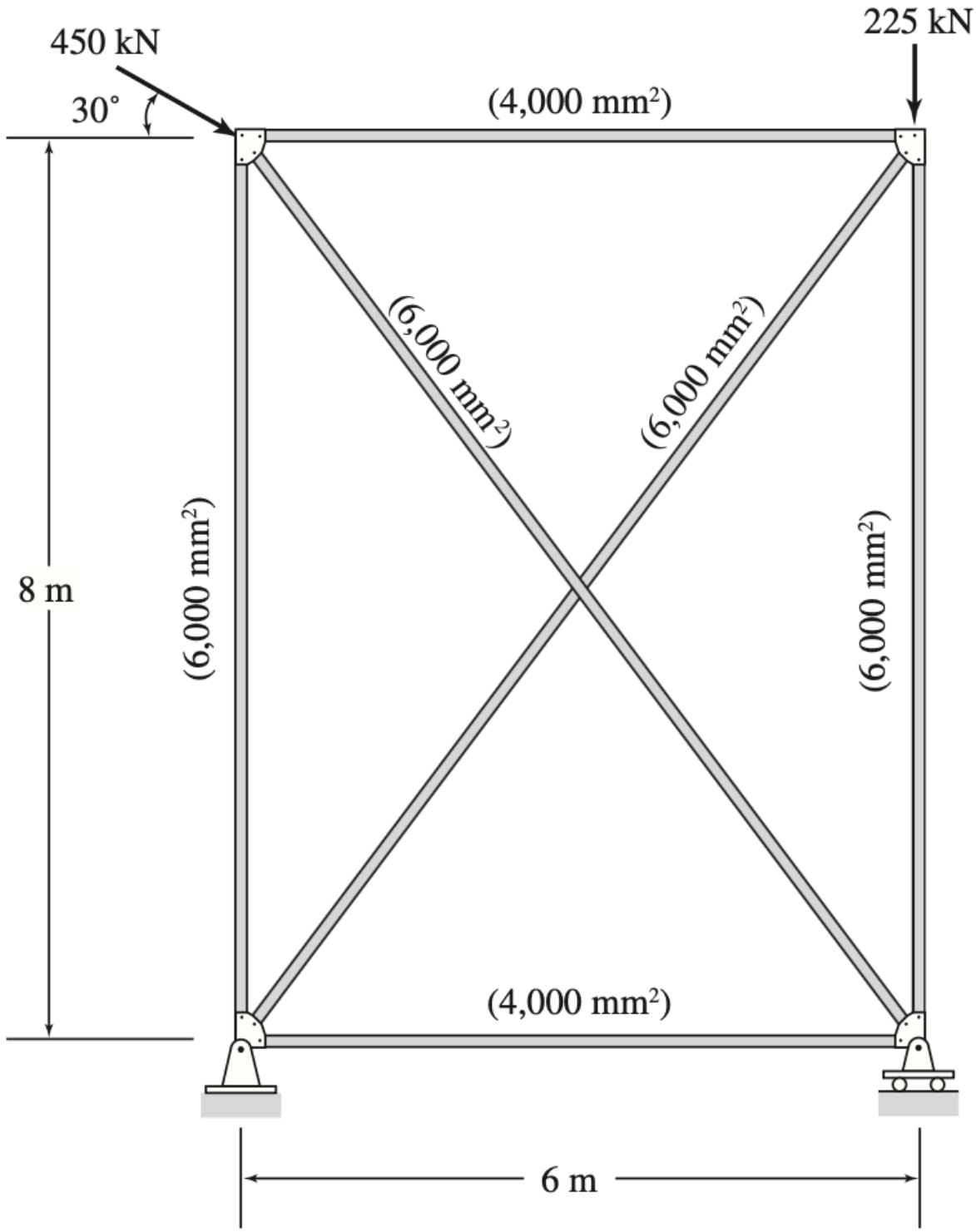
CEE6501 — Written Assignment, Week 3，Siyuan Wang



Question 1 — Structural Idealization

For the structure shown above:

1. Number the **joints (nodes)** on the figure.
2. Number the **members (elements)**.
3. Assign **global degree-of-freedom (DOF) numbers** at each node.
4. Identify the **DOF types** using the following notation:
    - Circle: free (unknown) DOFs
    - Box: DOFs with applied loads
    - Underline: restrained DOFs (reaction forces)
5. Write the **global displacement vector**, $\boldsymbol{u}$.
6. Write the **global force vector**, $\boldsymbol{f}$.

Solution for Question 1

1. Number the **joints (nodes)** on the figure.
2. Number the **members (elements)**.
3. Assign **global degree-of-freedom (DOF) numbers** at each node.
4. Identify the **DOF types** using the following notation:
    - Circle: free (unknown) DOFs
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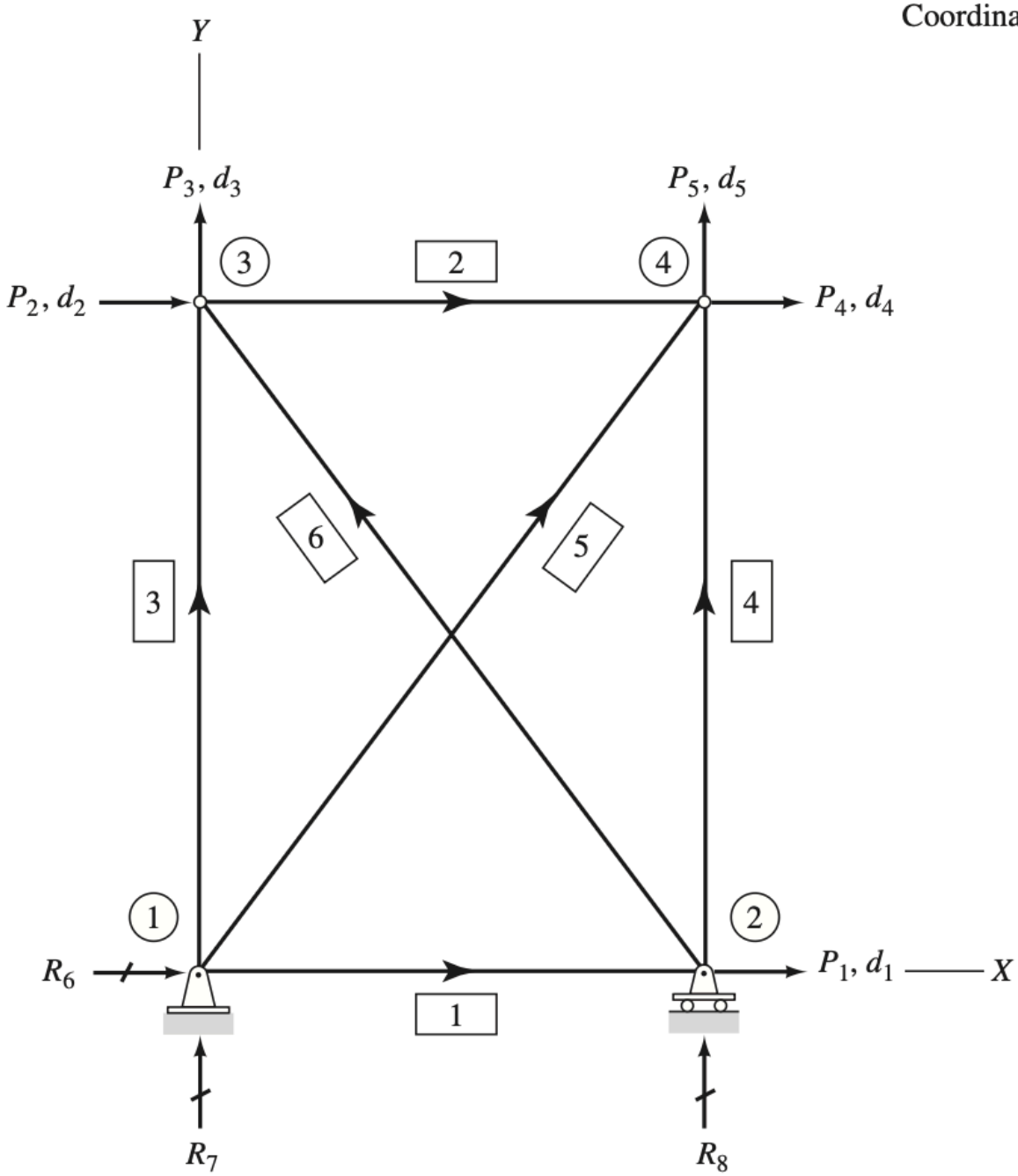

5. Write the **global displacement vector**, $\boldsymbol{u}$.

$$\mathbf{u} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

6. Write the **global force vector**, $\boldsymbol{f}$.

$$\mathbf{f} = \begin{bmatrix} 0 \\ 389.7 \\ -225 \\ 0 \\ -225 \\ R_6 \\ R_7 \\ R_8 \end{bmatrix} kN$$

Question 2 — Transformation Matrices

Calculate the **transformation matrix**, $\boldsymbol{T}$, for the following members:

1. **Diagonal member**
   Start node: top left
   End node: bottom right
2. **Vertical member**
   Start node: bottom left
   End node: top left
3. **Horizontal member**
   Start node: top right
   End node: top left

Note:
The angle, θ, is measured as the counter-clockwise rotation from the global x-axis (horizontal, positive to the right) to the local x-axis of the element.

Solution for Question 2

Calculate the **transformation matrix**, $\boldsymbol{T}$, for the following members:

1. **Diagonal member**

$$\sin\theta = -\frac{4}{5} \qquad \cos\theta = \frac{3}{5}$$

$$\boldsymbol{T} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & \cos\theta & \sin\theta \\ 0 & 0 & -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ 0 & 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

2. **Vertical member**

$$\sin\theta = 1 \qquad \cos\theta = 0$$

$$\boldsymbol{T} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & \cos\theta & \sin\theta \\ 0 & 0 & -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

3. **Horizontal member**

$$\sin\theta = 0 \qquad \cos\theta = -1$$

$$\boldsymbol{T} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & \cos\theta & \sin\theta \\ 0 & 0 & -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Question 3 — Global Stiffness Matrix

Compute the **global stiffness matrix** for all elements from Question 2.

- For the **diagonal member**, derive the global stiffness matrix using **matrix multiplication** and show that it matches the **closed-form expression** presented on **Slide 21 of Lecture 3.2**.
- For the remaining members, it is sufficient to present the **closed-form solutions** directly.

Solution for Question 3

1. **Diagonal member**

$$\boldsymbol{K} = \boldsymbol{T}^{\mathsf{T}} \boldsymbol{k} \boldsymbol{T}$$

$$\boldsymbol{K} = \frac{EA}{L} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} & 0 & 0 \\ -\frac{4}{5} & \frac{3}{5} & 0 & 0 \\ 0 & 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ 0 & 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$= \frac{EA}{L} \begin{bmatrix} \frac{9}{25} & -\frac{12}{25} & -\frac{9}{25} & \frac{12}{25} \\ -\frac{12}{25} & \frac{16}{25} & \frac{12}{25} & -\frac{16}{25} \\ -\frac{9}{25} & \frac{12}{25} & \frac{9}{25} & -\frac{12}{25} \\ \frac{12}{25} & -\frac{16}{25} & -\frac{12}{25} & \frac{16}{25} \end{bmatrix}$$

2. **Vertical member**

$$\boldsymbol{K} = \boldsymbol{T}^{\mathsf{T}} \boldsymbol{k} \boldsymbol{T}$$

$$\boldsymbol{K} = \frac{EA}{L} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$= \frac{EA}{L} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

3. **Horizontal member**

$$\boldsymbol{K} = \boldsymbol{T}^{\mathsf{T}} \boldsymbol{k} \boldsymbol{T}$$

$$\boldsymbol{K} = \frac{EA}{L} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$= \frac{EA}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Question 4 — Unit Displacement Method

Using the **unit displacement method** described on **Slide 23**, derive the **second column** of the global stiffness matrix for a truss element.

Solution for Question 4

- Impose $v_2 = 1$, all other global displacements zero

$$v_2 = 1, \qquad v_1 = v_3 = v_4 = 0$$

So:

$$\boldsymbol{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

- Project the resulting axial deformation onto the member axis

Axial displacement (along local x') at each node:

$$u'_i = v_1 \cos\theta + v_2 \sin\theta$$

$$u'_j = v_3 \cos\theta + v_4 \sin\theta$$

Substitute $v_2 = 1,\ v_1 = v_3 = v_4 = 0$:

$$u_i' = 0 \cdot \cos\theta + 1 \cdot \sin\theta = \sin\theta$$

$$u_j' = 0$$

Axial relative extension (positive in the $j - i$ direction):

$$\delta = u_j' - u_i' = 0 - \sin\theta = -\sin\theta$$

- Resolve the axial force back into global components

$$N = k\,\delta = \frac{EA}{L}(-\sin\theta) = -\frac{EA}{L}\sin\theta$$

Global nodal force at node i:

$$\boldsymbol{F_i} = N \begin{bmatrix} -\cos\theta \\ -\sin\theta \end{bmatrix}$$

Global nodal force at node j:

$$\boldsymbol{F_j} = N \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

With $N = -\dfrac{EA}{L}\sin\theta$, the force components are:

$$F_1 = \frac{EA}{L}\cos\theta\sin\theta$$

$$F_2 = \frac{EA}{L}\sin^2\theta$$

$$F_3 = -\frac{EA}{L}\cos\theta\sin\theta$$

$$F_4 = -\frac{EA}{L}\sin^2\theta$$

Therefore, for $\boldsymbol{v} = [0, 1, 0, 0]^T$:

$$\boldsymbol{F} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \frac{EA}{L} \begin{bmatrix} \cos\theta\sin\theta \\ \sin^2\theta \\ -\cos\theta\sin\theta \\ -\sin^2\theta \end{bmatrix}$$